

Online appendix for “Endogenous Plucking Through Networks: The Plucking Paradox”

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A Derivations for the static equilibrium

This appendix derives the static equilibrium conditions reported in Section 2. We assume perfect competition in the simple sector and monopolistic competition for varieties sold to the retailer. All firms are symmetric within each sector.

Marginal costs. The network firm minimizes the cost of the Cobb–Douglas aggregator in the main text. Let λ^N denote unit cost for a network firm and λ^S the unit cost for a simple firm. With competitive input prices and symmetry,

$$\ln \lambda^N = \alpha \ln \lambda^{N,\text{agg}} + (1 - \alpha) \ln \lambda^{S,\text{agg}} - \ln A, \quad (1)$$

$$\ln \lambda^S = \ln w - \ln A. \quad (2)$$

Prices. Equating $\lambda^S, \lambda^N, \alpha$ in (2) with $\lambda^{S,\text{agg}}, \lambda^{N,\text{agg}}, \alpha^{\text{agg}}$ since the model features representative firms, we can solve for the sectoral prices as functions of A and w

$$\ln \lambda^{N,\text{agg}} = \ln w - \frac{2 - \alpha^{\text{agg}}}{1 - \alpha^{\text{agg}}} \ln A, \quad (3)$$

$$\ln \lambda^{S,\text{agg}} = \ln w - \ln A. \quad (4)$$

Retailer’s cost minimization. The retailer faces constant-elasticity demand within each block, so the sectoral price indices satisfy

$$P = (p^{N,\text{agg}})^{\zeta} (p^{S,\text{agg}})^{1-\zeta} = \left(\frac{\sigma}{\sigma - 1} \right)^{\zeta} (\lambda^{N,\text{agg}})^{\zeta} (\lambda^{S,\text{agg}})^{1-\zeta}. \quad (5)$$

Normalizing $P \equiv 1$ and substituting (1)–(2) yields

$$\ln w = \left(\frac{\zeta}{1 - \alpha^{\text{agg}}} + 1 \right) \ln A - \zeta \ln \frac{\sigma}{\sigma - 1}, \quad (6)$$

which corresponds to the wage equation in the text.

Accounting and profits. Let TC denote total cost of network intermediates used by network firms. A fraction αTC is network goods and $(1 - \alpha)TC$ is simple goods. Total cost

of simple goods is wL , so the retailer spends $wL - (1 - \alpha)TC$ on simple goods and, by CES expenditure shares, $\frac{\zeta}{1-\zeta}(wL - (1 - \alpha)TC)$ on network goods. Converting retailer spending on network goods into costs using the sectoral markup implies

$$\alpha TC + \frac{\zeta}{1-\zeta} \frac{\sigma-1}{\sigma} (wL - (1 - \alpha)TC) = TC, \quad (7)$$

so

$$TC = \frac{\zeta(\sigma-1)}{(1-\alpha)(\sigma-\zeta)} wL. \quad (8)$$

Retailer spending on network goods is then $\frac{\sigma\zeta}{\sigma-\zeta}wL$. Network firms earn revenue from two sources: sales to the retailer at the markup price, and intermediate sales of αTC to other network firms at marginal cost. The latter carry no margin but exactly offset the network-goods fraction of input costs, so network profits equal total revenue minus total input cost,

$$\pi^N = \underbrace{\frac{\sigma\zeta}{\sigma-\zeta}wL}_{\text{retail sales}} + \underbrace{\alpha TC}_{\text{intermediate sales}} - \underbrace{TC}_{\text{input cost}} = \frac{\zeta}{\sigma-\zeta}wL, \quad (9)$$

while perfect competition in the simple sector implies

$$\pi^S = 0. \quad (10)$$

Finally, total consumption equals total revenue,

$$C = \frac{\sigma}{\sigma-\zeta}wL = wL + \pi^N, \quad (11)$$

which matches the aggregate consumption expression in the main text.

Firm-specific profits. Let $X = [\alpha^{\text{agg}}, A]$ and consider a firm that chooses its own network intensity α while taking aggregate prices as given. Using (1) with aggregate prices (3)–(4)

and (6), the firm's unit cost satisfies

$$\ln \lambda^N(\alpha; X) = \left(\frac{\zeta - \alpha}{1 - \alpha^{\text{agg}}} - 1 \right) \ln A - \zeta \ln \frac{\sigma}{\sigma - 1}, \quad (12)$$

The aggregate $\lambda^{N,\text{agg}}(X)$ satisfies

$$\ln \lambda^{N,\text{agg}}(X) = \left(\frac{\zeta - 1}{1 - \alpha^{\text{agg}}} \right) \ln A - \zeta \ln \frac{\sigma}{\sigma - 1}. \quad (13)$$

Under CES demand, revenue is proportional to $(p(\alpha; X)/p^{N,\text{agg}}(X))^{1-\sigma}$, implying

$$\pi^N(\alpha; X) = \pi^{N,\text{agg}}(X) \left(\frac{p(\alpha; X)}{p^{N,\text{agg}}(X)} \right)^{1-\sigma} \quad (14)$$

$$= \pi^{N,\text{agg}}(X) \left(\frac{\lambda^N(\alpha; X)}{\lambda^{N,\text{agg}}(X)} \right)^{1-\sigma} \quad (15)$$

$$= \frac{\zeta}{\sigma - \zeta} \left(\frac{\sigma - 1}{\sigma} \right)^\zeta A^{\frac{\zeta + (1 - (1 - \delta)\alpha)(1 - \sigma)}{1 - (1 - \delta)\alpha^{\text{agg}}} + \sigma} L, \quad (16)$$

which is consistent with the main text.

B Derivations for the optimal fiscal policy

This appendix derives the fiscal wedges that decentralize the planner allocation. Throughout this section, we work with the dynamic model: the effective network intensity entering the static block is $(1 - \delta)\alpha$, where α is the previous period's choice. The implementation requires two margins. A wage subsidy implements the planner's intratemporal labor condition, and a proportional subsidy on network profits implements the planner's Euler equation for network investment.

Static objects. Let $X = (\alpha, A)$ and let $\tilde{c}(X)$ denote the GHH consumption aggregate entering marginal utility. From Appendix A, the equilibrium wage in state X is

$$w(X) = \left(\frac{\sigma - 1}{\sigma} \right)^\zeta A^{\frac{\zeta}{1 - (1 - \delta)\alpha} + 1}. \quad (17)$$

Before imposing labor-market clearing, network profits at the symmetric allocation can be written as

$$\pi(\alpha, L; X) = \frac{\zeta}{\sigma - \zeta} \left(\frac{\sigma - 1}{\sigma} \right)^\zeta A^{\frac{\zeta}{1 - (1 - \delta)\alpha} + 1} L = \frac{\zeta}{\sigma - \zeta} w(X) L. \quad (18)$$

From the firm-specific profit expression in Appendix A, the private marginal profit from a firm's own network choice, holding aggregate prices and current labor fixed, is

$$\begin{aligned} \pi_1(\alpha, L; X) &= \pi(\alpha, L; X) \log(A) \frac{(\sigma - 1)(1 - \delta)}{1 - (1 - \delta)\alpha} \\ &= \frac{\zeta}{\sigma - \zeta} w(X) L \log(A) \frac{(\sigma - 1)(1 - \delta)}{1 - (1 - \delta)\alpha}. \end{aligned} \quad (19)$$

This is the object that enters the firm's Euler equation.

Labor implementation. With a wage subsidy τ^L , the household intratemporal condition under GHH preferences is

$$\eta n(X)^{1/\chi} = (1 + \tau^L) w(X). \quad (20)$$

At state $X = (\alpha, A)$, the planner's gross output as a function of labor n is

$$Y(\alpha, A, n) = \frac{\sigma}{\sigma - \zeta} \left(\frac{\sigma - 1}{\sigma} \right)^\zeta A^{\frac{\zeta}{1 - (1 - \delta)\alpha} + 1} n. \quad (21)$$

The planner's intratemporal condition is then

$$\eta n(X)^{1/\chi} = \left. \frac{\partial Y(\alpha, A, n)}{\partial n} \right|_{n=n(X)} = \frac{\sigma}{\sigma - \zeta} \left(\frac{\sigma - 1}{\sigma} \right)^\zeta A^{\frac{\zeta}{1 - (1 - \delta)\alpha} + 1} = \frac{\sigma}{\sigma - \zeta} w(X). \quad (22)$$

The SPP problem does not have a wage object. The last equality only uses the CE wage schedule in (17), evaluated at the same aggregate state X , to write the planner's marginal product of labor compactly as $\frac{\sigma}{\sigma - \zeta} w(X)$. Thus the wage subsidy that implements planner

labor is

$$1 + \tau^L = \frac{\sigma}{\sigma - \zeta}. \quad (23)$$

Under (23), the CE household FOC (20) coincides with the planner's intratemporal condition (22). Hence, conditional on the same aggregate state X , the implemented CE labor supply equals planner labor: $n^{CE}(X) = n^{SPP}(X)$.

Planner's Euler equation. The planner's marginal resource benefit from network intensity is the partial derivative of output with respect to α , evaluated at planner labor. Following (22), we write the planner's marginal product of labor as $\frac{\sigma}{\sigma - \zeta}w(X)$, where $w(X)$ is the CE wage schedule evaluated at the same aggregate state X . Therefore,

$$\left. \frac{\partial Y(\alpha, A, n)}{\partial \alpha} \right|_{n=n^{SPP}(X)} = \frac{\sigma}{\sigma - \zeta}w(X)n^{SPP}(X)\log(A)\frac{\zeta(1 - \delta)}{(1 - (1 - \delta)\alpha)^2}. \quad (24)$$

Ignoring corner multipliers, the planner Euler equation is

$$\mathbb{E} \left[\beta \frac{(\tilde{c}'(X'))^{-\rho}}{(\tilde{c}(X))^{-\rho}} \left(\left. \frac{\partial Y(\alpha', A', n')}{\partial \alpha'} \right|_{n'=n^{SPP}(X')} - \Phi_1(\alpha', \alpha'') \right) \middle| X \right] = \Phi_2(\alpha, \alpha'). \quad (25)$$

Network-profit subsidy. With planner labor implemented, the firm Euler is

$$\mathbb{E} \left[\beta \frac{(\tilde{c}'(X'))^{-\rho}}{(\tilde{c}(X))^{-\rho}} ((1 + \tau(\alpha'))\pi_1(\alpha', n^{SPP}(X'); X') - \Phi_1(\alpha', \alpha'')) \middle| X \right] = \Phi_2(\alpha, \alpha'). \quad (26)$$

Equating the marginal return terms in (25) and (26), and using (19) and (24), gives

$$\begin{aligned} 1 + \tau(\alpha') &= \frac{\left. \frac{\partial Y(\alpha', A', n')}{\partial \alpha'} \right|_{n'=n^{SPP}(X')}}{\pi_1(\alpha', n^{SPP}(X'); X')} \\ &= \frac{\frac{\sigma}{\sigma - \zeta}w(X')n^{SPP}(X')\log(A')\frac{\zeta(1 - \delta)}{(1 - (1 - \delta)\alpha')^2}}{\frac{\zeta}{\sigma - \zeta}w(X')n^{SPP}(X')\log(A')\frac{(\sigma - 1)(1 - \delta)}{1 - (1 - \delta)\alpha'}} \\ &= \frac{\sigma}{\sigma - 1} \frac{1}{1 - (1 - \delta)\alpha'}. \end{aligned} \quad (27)$$

The profit subsidy is therefore the network Euler wedge conditional on planner labor.

For comparison, suppose there is no wage subsidy. Let $n^{CE,0}(X)$ denote CE labor in that case. From (20) with $\tau^L = 0$ and the planner intratemporal condition (22),

$$\frac{n^{SPP}(X)}{n^{CE,0}(X)} = \left(\frac{\sigma}{\sigma - \zeta} \right)^\chi. \quad (28)$$

If one forces the network subsidy alone to equate the planner marginal benefit to the private marginal profit term, the implied subsidy is

$$\begin{aligned} 1 + \tau^0(\alpha') &= \frac{\left. \frac{\partial Y(\alpha', A', n')}{\partial \alpha'} \right|_{n'=n^{SPP}(X')}}{\pi_1(\alpha', n^{CE,0}(X'); X')} \\ &= \frac{\sigma}{\sigma - 1} \frac{1}{1 - (1 - \delta)\alpha'} \left(\frac{\sigma}{\sigma - \zeta} \right)^\chi. \end{aligned} \quad (29)$$

The extra factor is exactly the missing labor wedge. This is the formula one obtains if the corporate subsidy is the only fiscal instrument and the network marginal-return comparison is made using decentralized labor. Because that labor allocation also changes consumption and the SDF, this corporate-only formula does not generally recover the planner Euler equation or the planner labor allocation.

Budget balance. The wage subsidy and the profit subsidy are financed by lump-sum taxes. Government budget balance requires

$$T(X) = \tau^L w(X) n(X) + \tau(\alpha) \pi(\alpha, L(X); X), \quad L(X) = n(X). \quad (30)$$

Using

$$c = (1 + \tau^L) w(X) n(X) + (1 + \tau(\alpha)) \pi(\alpha, L(X); X) - \Phi(\alpha, \alpha') - T(X), \quad (31)$$

the aggregate resource constraint reduces to

$$c = w(X) L(X) + \pi(\alpha, L(X); X) - \Phi(\alpha, \alpha'). \quad (32)$$

Thus the subsidies implement wedges between private and social marginal conditions without changing aggregate resources.

C A plucking-cycle exercise in simulated output

This appendix implements an analogous plucking-cycle exercise in the simulated model, using the threshold turning-point algorithm in [Dupraz, Nakamura, and Steinsson \(2025\)](#). The exercise asks whether the model-generated output series displays the one-sided predictability pattern emphasized in the plucking literature: contraction amplitudes predict subsequent recoveries, while recovery amplitudes have little predictive power for subsequent contractions.

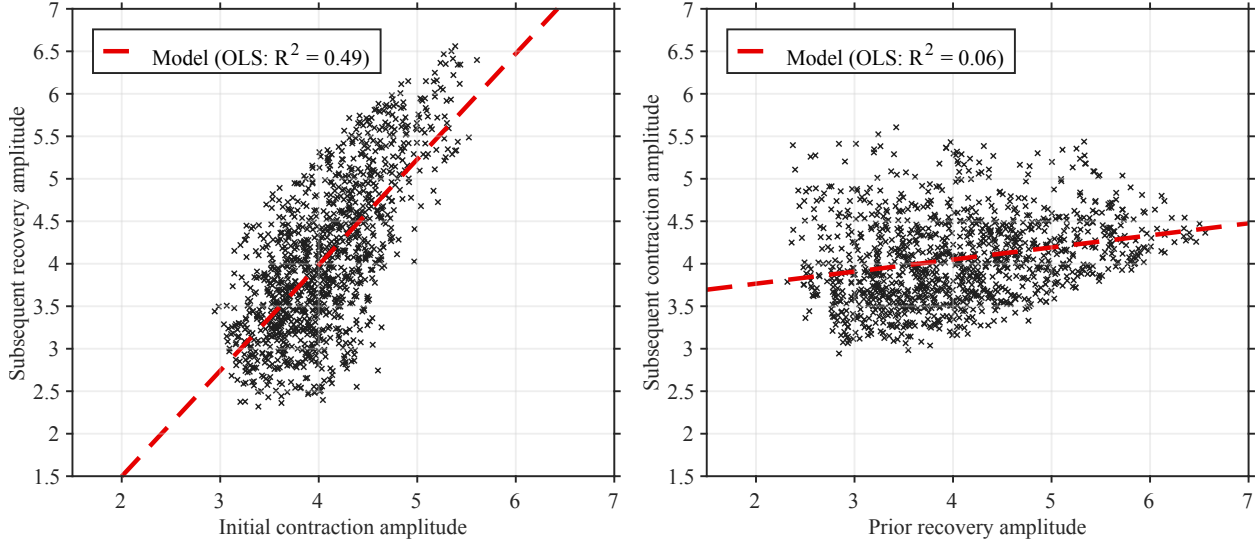
The empirical implementation in [Dupraz, Nakamura, and Steinsson \(2025\)](#) identifies peaks and troughs in unemployment in levels. Since the algorithm is designed for unemployment, where higher values indicate greater slack, we transform simulated output into a comparable shortfall measure. Consistent with Friedman’s ceiling interpretation, we measure output relative to a smooth moving ceiling and define

$$u_t = -\hat{y}_t, \quad \hat{y}_t = \log Y_t - \text{HP}(\log Y_t),$$

where the HP smoothing parameter is $\lambda = 6.25$ for annual data. We then apply the threshold turning-point logic to u_t , with the cycle-size threshold set to 1.5 standard deviations of u_t . For each identified cycle, we compute the initial contraction amplitude and the subsequent recovery amplitude from the output-gap series $\hat{y}_t$.

Figure [C.1](#) reports the two plucking regressions. The left panel regresses the subsequent recovery amplitude on the initial contraction amplitude. The right panel regresses the subsequent contraction amplitude on the prior recovery amplitude. In the simulated network economy, contractions strongly predict subsequent recoveries, with an R^2 of 0.49. In contrast, prior recoveries explain little of the next contraction, with an R^2 of 0.06. This one-sided predictability is consistent with the model’s plucking interpretation: downturns represent stochastic drops below a moving ceiling, while recoveries reflect within-regime convergence back toward that ceiling. Hence recovery is tied to the size of the inherited shortfall, whereas

Figure C.1: Plucking-cycle exercise for simulated output slack



Notes: The figure applies the turning-point algorithm in [Dupraz, Nakamura, and Steinsson \(2025\)](#) to simulated model output slack, $u_t = -\hat{y}_t$, where $\hat{y}_t$ is the HP-filtered log output gap. The cycle-size threshold is 1.5 standard deviations of u_t . Amplitudes are expressed as percent changes in output relative to the smooth ceiling. The left panel regresses subsequent recovery amplitudes on initial contraction amplitudes. The right panel regresses subsequent contraction amplitudes on prior recovery amplitudes.

the next downturn is governed by a new aggregate-state realization rather than mechanically by the size of the previous recovery.

D Additional empirical results

In this section, we present additional empirical results that further support the model's mechanism and its implications for sectoral employment dynamics during recessions. Table [D.2](#) reports the results from the regression of the depth of employment contraction on material share (with energy cost included), controlling for various industry characteristics. The coefficient on material share remains positive and statistically significant across all specifications, reinforcing the finding that industries with higher material intensity experience deeper contractions during recessions.

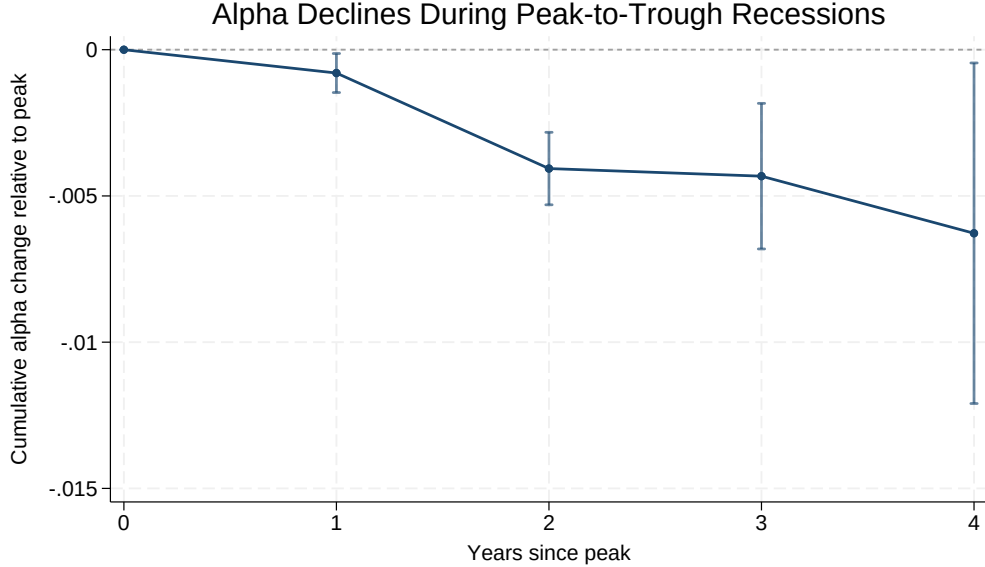
Table [D.3](#) presents the results from the trough-level regressions, showing that industries with higher material share (with energy cost included) not only experience deeper contractions but also reach lower levels of employment and total payroll at the trough. This further

Table D.1: Peak-to-trough recessions and material share adjustment

	(1)	(2)
	No FE	Industry FE
Recession	-0.0029*** (0.0004)	-0.0026*** (0.0004)
Industry FE	No	Yes
R-squared	0.003	0.074
N	21205	21205

Notes: The dependent variable is $\Delta\alpha_{it} = \alpha_{it} - \alpha_{i,t-1}$, where $\alpha_{it} = \text{matcost}_{it}/\text{vship}_{it}$. Recession is an indicator for annual transitions that lie between an industry-level employment-cycle peak and the next trough. Employment cycles are computed using an annual HP filter with smoothing parameter 6.25. Standard errors are clustered by industry. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure D.2: Material share during peak-to-trough recessions



Notes: The figure plots the average cumulative change in material share, $\alpha_{it} - \alpha_{i,\text{peak}}$, during industry-level recession episodes. Peaks and troughs are local turning points in HP-filtered employment, using the annual smoothing parameter 6.25. Vertical bars report pointwise 95 percent confidence intervals.

corroborates the model's mechanism, as a more network-intensive production structure exacerbates the depth of downturns, leading to more pronounced declines in key economic indicators during recessions.

We also examine whether material intensity adjusts over the cycle. We proxy network intensity by the material cost share, $\alpha_{it} = \text{matcost}_{it}/\text{vship}_{it}$, and define recession transitions

Table D.2: Material share (including energy) and peak-to-trough drop [Data]

Drop ^Y , Y=	(1) emp	(2) pay	(3) emp	(4) pay	(5) emp	(6) pay
Material Cost Share	0.132 (0.039)	0.140 (0.042)			0.125 (0.041)	0.123 (0.043)
Booming Duration			0.014 (0.002)	0.011 (0.002)	0.014 (0.002)	0.011 (0.002)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Peak year FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.349	0.352	0.363	0.365	0.365	0.367
N	5882	5882	5518	5518	5518	5518

Notes: Dependent variable is the peak-to-trough drop in employment, payroll, value added. Material share is measured at the peak year (pre-determined). Industry and peak-year fixed effects are included. Capital stock and the outcome variable at the peak level is included as control; standard errors are clustered by industry. Standard errors in parentheses.

as years that lie between an industry-level employment-cycle peak and the next trough. Table D.1 shows that material share growth is lower during these peak-to-trough recessions, both without fixed effects and after absorbing industry fixed effects. Figure D.2 plots the corresponding average path of material share relative to its peak value during recession episodes.

E Canonical RBC model for benchmark

This appendix describes the canonical RBC model used to construct the benchmark conditional saddle paths in Figure 5(a). Throughout this appendix, α denotes the capital share and δ the capital depreciation rate, following standard RBC convention; these symbols are distinct from the network intensity and the network depreciation rate of the main text. A

Table D.3: Trough level regression (including energy cost in material cost) [Data]

Drop ^Y , Y=	(1) emp	(2) pay	(3) emp	(4) pay	(5) emp	(6) pay
Material Cost Share	-0.131 (0.039)	-0.141 (0.043)			-0.122 (0.041)	-0.123 (0.044)
Booming Duration			-0.014 (0.002)	-0.011 (0.002)	-0.014 (0.002)	-0.011 (0.002)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Peak year FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.989	0.992	0.989	0.991	0.989	0.991
N	5882	5882	5518	5518	5518	5518

Notes: Dependent variable is the drop in employment, payroll. Material share is measured at the peak year (pre-determined). Industry and peak-year fixed effects are included. Capital stock at the peak level is included as control; standard errors are clustered by industry. Standard errors in parentheses.

representative household solves

$$v(a; K, A) = \max_{c, a'} \log(c) + \beta \mathbb{E}[v(a'; K', A') \mid A] \quad (33)$$

$$\text{s.t. } c + a' = a(1 + r(K, A)) + w(K, A) \quad (34)$$

$$a' \geq 0, \quad (35)$$

where v is the value function, a is wealth at the beginning of the period, K is aggregate capital, and A is aggregate TFP. The aggregate state evolves according to

$$K' = \Gamma_{\text{endo}}(K, A), \quad A' \sim \Gamma_{\text{exo}}(\cdot \mid A), \quad (36)$$

where Γ_{endo} is the equilibrium law of motion for capital. TFP follows a two-state Markov chain between $A \in \{B, G\}$ with $G > B$ and transition matrix

$$\Gamma_{\text{exo}} = \begin{bmatrix} \pi_{BB} & \pi_{BG} \\ \pi_{GB} & \pi_{GG} \end{bmatrix}. \quad (37)$$

Production is Cobb-Douglas with constant returns to scale. Competitive factor prices are

$$r(K, A) = \alpha AK^{\alpha-1} - \delta, \quad (38)$$

$$w(K, A) = (1 - \alpha)AK^\alpha, \quad (39)$$

where δ is the depreciation rate. In equilibrium, the representative household's asset holdings equal the aggregate capital stock: $a = K$.

F Robustness of the fiscal wedges without GHH

The GHH specification is useful because it removes wealth effects from labor supply and gives a closed-form labor rule. The two-wedge implementation above does not rely on that simplification. To see this, suppose instead that household preferences are separable:

$$u(c, n) = \frac{c^{1-\rho}}{1-\rho} - \eta \frac{n^{1+1/\chi}}{1+1/\chi}.$$

With a wage subsidy, the household intratemporal condition is

$$\eta n(X)^{1/\chi} = c(X)^{-\rho}(1 + \tau^L)w(X). \quad (40)$$

The planner intratemporal condition is

$$\eta n(X)^{1/\chi} = c(X)^{-\rho} \frac{\partial Y(\alpha, A, n)}{\partial n} = c(X)^{-\rho} \frac{\sigma}{\sigma - \zeta} w(X). \quad (41)$$

Thus, at the implemented allocation, the same constant wage subsidy aligns the labor margin:

$$1 + \tau^L = \frac{\sigma}{\sigma - \zeta}.$$

The difference from GHH is not the size of the labor wedge, but the form of labor supply. With separable CRRA preferences, labor depends on consumption through $c(X)^{-\rho}$, so labor and consumption must be solved jointly with the household budget constraint and the dynamic Euler equation.

Conditional on implementing planner labor, the network-profit subsidy is unchanged. The planner marginal resource benefit and the private marginal profit term are the same objects as in (24) and (19), evaluated at the implemented labor allocation. Therefore their ratio remains

$$1 + \tau(\alpha') = \frac{\sigma}{\sigma - 1} \frac{1}{1 - (1 - \delta)\alpha'}.$$

Without the wage subsidy, a corporate subsidy alone does not directly align the household intratemporal condition. Under separable CRRA preferences this also changes consumption and the SDF, so there is no simple state-only corporate subsidy formula that fully decentralizes the planner allocation.

G CES extensions for unit costs

This appendix checks how the static sign result

$$\frac{\partial \ln \lambda^N}{\partial \ln A} < 0, \quad \frac{\partial \ln \lambda^S}{\partial \ln A} > 0$$

extends beyond Cobb–Douglas aggregation. Throughout, hold aggregate network intensity fixed at $\alpha \in (0, 1)$ and normalize the final-good price index to one. The Cobb–Douglas cases $\varepsilon_F = 1$ and $\varepsilon_N = 1$ are understood as limits.

CES final sector. First keep the benchmark network-sector unit-cost equation:

$$\ln \lambda^N = \alpha \ln \lambda^N + (1 - \alpha) \ln \lambda^S - \ln A.$$

Then

$$\frac{\lambda^N}{\lambda^S} = A^{-1/(1-\alpha)}.$$

Now let the final sector be CES. The network-sector markup enters through the sector price $(\sigma/(\sigma - 1))\lambda^N$:

$$1 = \left[\zeta \left(\frac{\sigma}{\sigma - 1} \lambda^N \right)^{1-\varepsilon_F} + (1 - \zeta)(\lambda^S)^{1-\varepsilon_F} \right], \quad \zeta \in (0, 1). \quad (42)$$

Substituting the relative-cost equation into (42) gives

$$1 = \left[\zeta \left(\frac{\sigma}{\sigma - 1} \right)^{1-\varepsilon_F} A^{-(1-\varepsilon_F)/(1-\alpha)} + 1 - \zeta \right] (\lambda^S)^{1-\varepsilon_F}.$$

Therefore,

$$\begin{aligned} \frac{\partial \ln \lambda^S}{\partial \ln A} &= \frac{1}{1 - \alpha} \frac{\zeta \left(\frac{\sigma}{\sigma - 1} \right)^{1-\varepsilon_F} A^{-(1-\varepsilon_F)/(1-\alpha)}}{\zeta \left(\frac{\sigma}{\sigma - 1} \right)^{1-\varepsilon_F} A^{-(1-\varepsilon_F)/(1-\alpha)} + 1 - \zeta}, \\ \frac{\partial \ln \lambda^N}{\partial \ln A} &= -\frac{1}{1 - \alpha} \frac{1 - \zeta}{\zeta \left(\frac{\sigma}{\sigma - 1} \right)^{1-\varepsilon_F} A^{-(1-\varepsilon_F)/(1-\alpha)} + 1 - \zeta}. \end{aligned} \quad (43)$$

Evaluated at $A = 1$,

$$\begin{aligned} \left. \frac{\partial \ln \lambda^S}{\partial \ln A} \right|_{A=1} &= \frac{1}{1 - \alpha} \frac{\zeta \left(\frac{\sigma}{\sigma - 1} \right)^{1-\varepsilon_F}}{\zeta \left(\frac{\sigma}{\sigma - 1} \right)^{1-\varepsilon_F} + 1 - \zeta} > 0, \\ \left. \frac{\partial \ln \lambda^N}{\partial \ln A} \right|_{A=1} &= -\frac{1}{1 - \alpha} \frac{1 - \zeta}{\zeta \left(\frac{\sigma}{\sigma - 1} \right)^{1-\varepsilon_F} + 1 - \zeta} < 0. \end{aligned} \quad (44)$$

Thus the final-sector curvature changes the expenditure-share term away from $A = 1$, but not the signs. In this case the sign result is global for $A > 0$.

CES network sector. Now keep the final sector Cobb–Douglas:

$$1 = \left(\frac{\sigma}{\sigma - 1} \lambda^N \right)^\zeta (\lambda^S)^{1-\zeta}.$$

Let the network-sector unit cost be CES, with A entering as a level Hicks-neutral productivity shifter:

$$\lambda^N = \frac{1}{A} [\alpha(\lambda^N)^{1-\varepsilon_N} + (1 - \alpha)(\lambda^S)^{1-\varepsilon_N}]^{1/(1-\varepsilon_N)}.$$

Raising both sides to $1 - \varepsilon_N$ and rearranging gives

$$\left(\frac{\lambda^N}{\lambda^S}\right)^{1-\varepsilon_N} = \frac{1-\alpha}{A^{1-\varepsilon_N}-\alpha}.$$

This expression also makes clear the admissible region for A under the CES network-sector unit cost. Since $\lambda^N/\lambda^S > 0$, the denominator on the right-hand side must be positive:

$$A^{1-\varepsilon_N} > \alpha.$$

This restriction is automatically satisfied at $A = 1$. Away from $A = 1$, it can bind: when $\varepsilon_N < 1$, it rules out sufficiently low A ; when $\varepsilon_N > 1$, it rules out sufficiently high A . We therefore interpret this exercise locally around $A = 1$, or on any support of A contained in the admissible region. For A satisfying this condition,

$$\frac{\partial \ln(\lambda^N/\lambda^S)}{\partial \ln A} = -\frac{A^{1-\varepsilon_N}}{A^{1-\varepsilon_N}-\alpha}.$$

The Cobb–Douglas final-sector normalization implies

$$d \ln \lambda^S = -\zeta d \ln(\lambda^N/\lambda^S), \quad d \ln \lambda^N = (1 - \zeta) d \ln(\lambda^N/\lambda^S).$$

Therefore

$$\begin{aligned} \frac{\partial \ln \lambda^S}{\partial \ln A} &= \zeta \frac{A^{1-\varepsilon_N}}{A^{1-\varepsilon_N}-\alpha}, \\ \frac{\partial \ln \lambda^N}{\partial \ln A} &= -(1 - \zeta) \frac{A^{1-\varepsilon_N}}{A^{1-\varepsilon_N}-\alpha}. \end{aligned} \tag{45}$$

At $A = 1$,

$$\left. \frac{\partial \ln \lambda^S}{\partial \ln A} \right|_{A=1} = \frac{\zeta}{1-\alpha} > 0, \quad \left. \frac{\partial \ln \lambda^N}{\partial \ln A} \right|_{A=1} = -\frac{1-\zeta}{1-\alpha} < 0.$$

Thus, with a standard Hicks-neutral productivity shifter in the CES unit-cost function, the sign result is unchanged for both complements and substitutes locally around $A = 1$.

References

Dupraz, Stéphane, Emi Nakamura, and Jón Steinsson. 2025. “A plucking model of business cycles.” *Journal of Monetary Economics* 152:103766.